ceptible watermarks into AI-generated media, including audio outputs.

Autonomous AI gives developers extraordinary velocity, but it also demands machine-speed defence. We see our role as co-builders of shared technical infrastructure that equips defenders across the ecosystem. Earlier this year, we introduced open tools and standards designed to help defenders move faster and secure inter-agent interactions. We brought Sec-Gemini v3, our specialised cybersecurity agent that automates incident response, digital forensics and malware analysis in real time, to trusted Indian government and enterprise testers. Our CodeMender agent strengthens code security early in development by automatically writing and contributing security patches back to open-source software.

To secure increasingly autonomous systems, we follow our AI Control Roadmap, a framework designed to prevent sandbox escapes and unauthorised lateral movement by treating untrusted AI agents as potential "insider threats". As part of this defence-in-depth approach, we open-sourced CAPSEM (Capabilities Security for Agents) to isolate autonomous agents within virtual machines, while advancing open standards such as Device Bound Session Credentials to make stolen session cookies useless.

Technical guardrails alone are insufficient without a clear division of operational responsibility. No single laboratory or corporate entity should hold unilateral authority over the boundaries of AI.

Chairman and co-founder of Google DeepMind, Sir Demis Hassabis, has recently proposed an industry-wide standards body to establish clear standards and guardrails for advanced AI models. This builds on our track record of active collaboration, including co-founding the Frontier Model Forum in 2023 to address significant risks to public safety and develop common best practices.

Governance should be proportioned across the AI value chain. Developers own the base: baseline safety, rigorous evaluations and transparent documentation. Deployers and sectoral regulators own the ceiling: the context-specific safeguards for domains such as emergency triage, agrarian credit and prudential banking, which a model developer cannot calibrate. This keeps humans in control of critical workflows and avoids duplicate or disproportionate compliance burdens.

In a similar vein, AI model development and evaluation cannot be isolated to laboratory conditions. Ensuring technology is genuinely useful and reliable requires grounding models in local realities, regional contexts and representative datasets.

In India, we are collaborating with the Center for Responsible AI (CeRAI) at IIT Madras and MLCommons to develop the Hindi-language AILuminate Safety Benchmark. Through our Amplify Initiative, we are contributing to the AI Safety Commons to build diverse regional datasets and partnering with CeRAI (Centre for Responsible AI) to drive adoption of the Secure AI Framework across Indian industry.

Crucially, this ecosystem-first approach translates into tangible human impact. Clinical teams at AIIMS are adapting MedGemma to address local needs across dermatology, outpatient triaging and leprosy screening. Project Vaani, our collaboration with IISc Bangalore that has open-sourced over 31,000 hours of speech data across over 100 Indic languages, is enabling local innovators such as mWire Labs to train accurate voice models for underrepresented languages like Garo.

In agriculture, our AnthroKrishi team's agricultural AI models are helping partners such as the Government of Telangana's ADeX platform deliver targeted crop advisories.

Ultimately, AI's trajectory and long-term public trust in the technology's capabilities is defined by scientific discipline and responsible engineering.

By embedding safeguards into foundational architectures, aligning with global frameworks and co-creating with local domain experts, we ensure that as technology advances, human purpose remains firmly in the driver's seat. Because the race that matters is getting it right for society: creating AI that remains transparent, helpful and empowering for everyone.■

The author is Senior Director, Research, Google DeepMind

TAKEAWAYS

- Safety, privacy, red-teaming and content authentication should be integrated into AI models from the design stage onwards
- Developers are responsible for baseline model safety, while deployers and sector-specific regulators should manage real-world risks
- As AI agents become more capable, new technical guardrails and control mechanisms are needed to prevent misuse
- AI systems are more useful when developed with regional datasets and community needs in mind, as in healthcare and agriculture projects